



IT-SIMPLERA INSTITUTE

Red Team Cyber Security Internship Program

WEEK 02 ASSIGNMENT REPORT

Computer Networking Fundamentals and Network Configuration

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LinkedIn Post: <https://lnkd.in/p/dimaeC5R>

GitHub Repository: https://github.com/devAwais3/Week02_redTeam-internship-ITSimpleraInstitute

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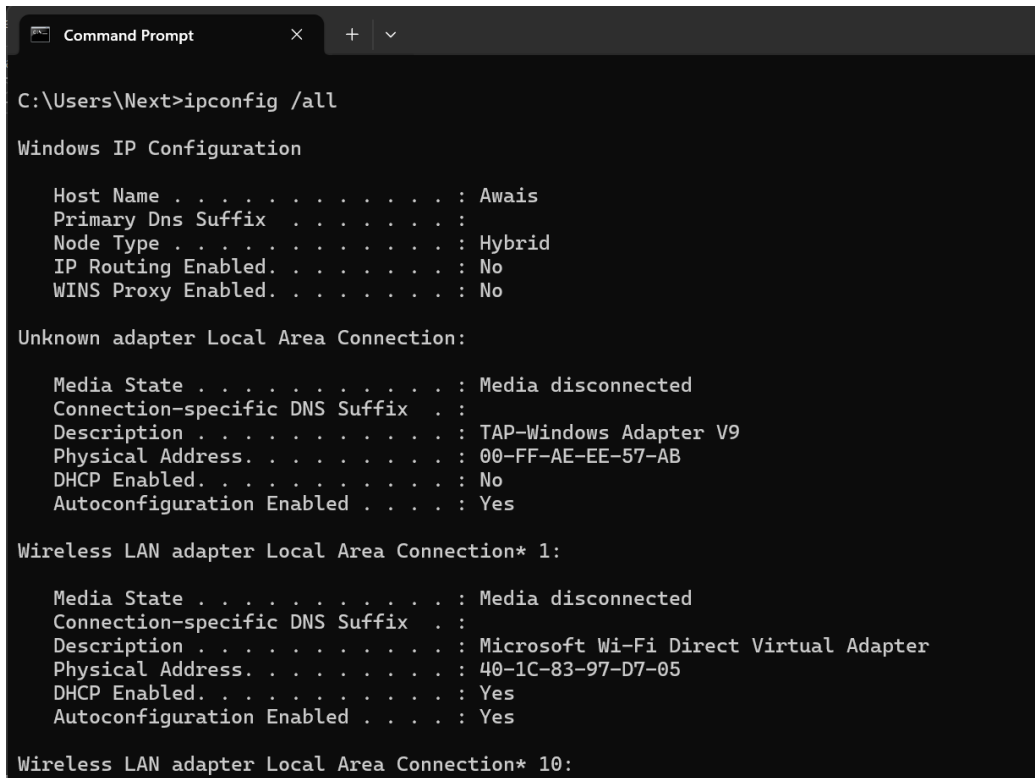
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1. Introduction

This report covers Week 02 of the Cyber Security Analyst Internship Program at IT-SIMPLERA Institute. It introduces core computer networking concepts including network configuration identification, essential networking commands, router versus switch comparison, IPv4 versus IPv6 addressing, and the seven layers of the OSI Model.

2. Task 1: Network Information

Command used: ipconfig /all



```
C:\Users\Next>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Awais
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Unknown adapter Local Area Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-AE-EE-57-AB
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 40-1C-83-97-D7-05
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:
```

```
Command Prompt
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 184569942
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-3C-80-D4-50-81-40-72-4F-1F
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wireless-AC 9560 160MHz
Physical Address. . . . . : 40-1C-83-97-D7-04
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2400:adcc:3101:4b00::1(Preferred)
Lease Obtained. . . . . : Saturday, July 11, 2026 8:23:38 PM
Lease Expires . . . . . : Sunday, July 12, 2026 8:23:38 PM
IPv6 Address. . . . . : 2400:adcc:3101:4b00:45cf:aa85:e9d8:7116(Preferred)
Temporary IPv6 Address. . . . . : 2400:adcc:3101:4b00:c5e8:7e06:2d85:df21(Preferred)
Link-local IPv6 Address . . . . . : fe80::24e:d9c0:92f8:e010%5(Preferred)
IPv4 Address. . . . . : 192.168.18.89(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, July 11, 2026 8:23:42 PM
Lease Expires . . . . . : Saturday, July 11, 2026 10:23:54 PM
Default Gateway . . . . . : fe80::1%5
                          192.168.18.1
DHCP Server . . . . . : 192.168.18.1
DHCPv6 IAID . . . . . : 71310467
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-3C-80-D4-50-81-40-72-4F-1F
DNS Servers . . . . . : fe80::1%5
                          192.168.18.1
```

Figure 1: Output of ipconfig /all showing adapter details, IPv4 address, and default gateway

IPv4 Address: 192.168.18.89

MAC (Physical) Address: 40-1C-83-97-D7-04

Default Gateway: 192.168.18.1

Purpose of each:

- IPv4 Address: identifies the device uniquely on the local network, allowing it to send and receive data.
- MAC Address: a hardware level physical address burned into the network adapter, used for communication within the local network segment.
- Default Gateway: the router address that the device uses to send traffic outside the local network, such as to the internet.

3. Task 2: Basic Networking Commands

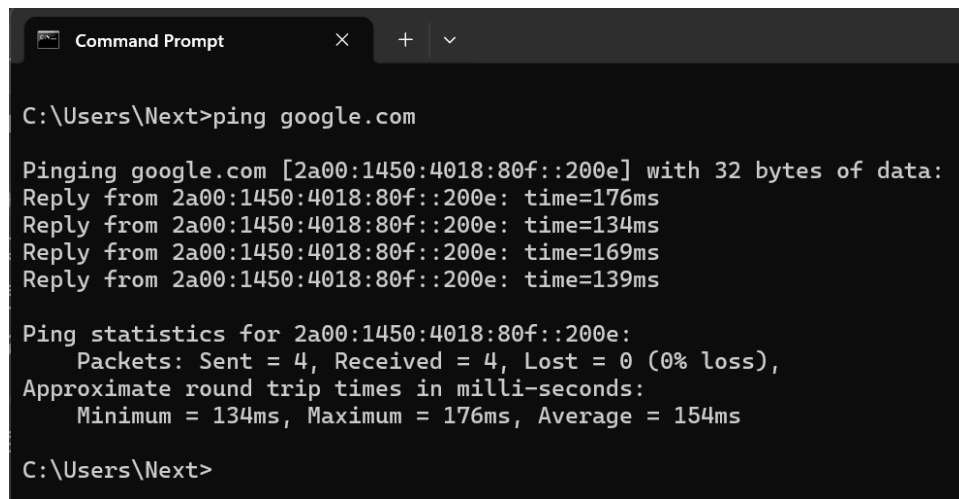
3.1 ipconfig /all

Purpose:

Displays complete network configuration details for all adapters, including IP address, subnet mask, gateway, and DNS servers.

Refer to Figure 1 in Task 1 above for the full output of this command.

3.2 ping google.com



```
C:\Users\Next>ping google.com

Pinging google.com [2a00:1450:4018:80f::200e] with 32 bytes of data:
Reply from 2a00:1450:4018:80f::200e: time=176ms
Reply from 2a00:1450:4018:80f::200e: time=134ms
Reply from 2a00:1450:4018:80f::200e: time=169ms
Reply from 2a00:1450:4018:80f::200e: time=139ms

Ping statistics for 2a00:1450:4018:80f::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 134ms, Maximum = 176ms, Average = 154ms

C:\Users\Next>
```

Figure 2: Output of ping google.com showing reply times and packet statistics

Purpose:

Tests connectivity to a remote host and measures round trip response time to check if the network path is reachable.

3.3 tracert google.com

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Next>tracert google.com

Tracing route to google.com [2a00:1450:4018:80f::200e]
over a maximum of 30 hops:

  1    2 ms    1 ms    1 ms  2400:adcc:3000:14b:21f5:a1c3:65c9:e0a2
  2    4 ms    2 ms    2 ms  2001:4538:e003:f000::1
  3    9 ms    8 ms    6 ms  2001:4538:0:e9::1
  4   66 ms   26 ms   37 ms  2001:4538:0:eb::1
  5   26 ms   27 ms   27 ms  2001:4538:a001:c5::1
  6   48 ms   46 ms   88 ms  2001:4538:bb:1::2
  7  131 ms  154 ms  162 ms  2400:adcf:ab::b
  8  188 ms  202 ms  139 ms  2a00:1450:84c2:40::1
  9  188 ms  159 ms  201 ms  2001:4860:0:1::92c
 10  168 ms  192 ms  139 ms  2001:4860:0:1::7d9b
 11  183 ms  162 ms  162 ms  lcmta-ah-in-x0e.1e100.net [2a00:1450:4018:80f::200e]

Trace complete.

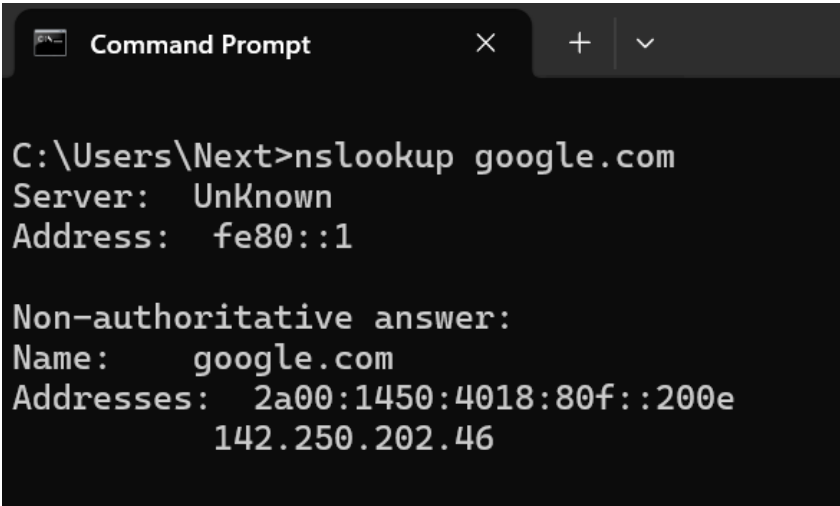
C:\Users\Next>
```

Figure 3: Output of tracert google.com showing each hop to the destination

Purpose:

Traces the route packets take to reach the destination, showing every hop along the path, useful for diagnosing where a connection slows down or fails.

3.4 nslookup google.com



```
Command Prompt

C:\Users\Next>nslookup google.com
Server:    UnKnown
Address:   fe80::1

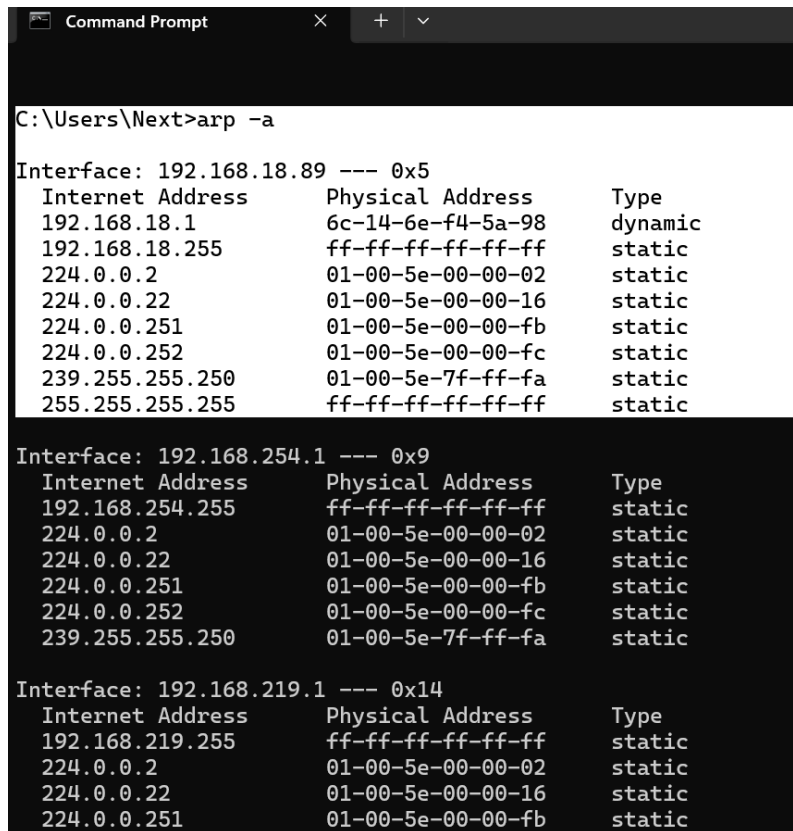
Non-authoritative answer:
Name:      google.com
Addresses: 2a00:1450:4018:80f::200e
           142.250.202.46
```

Figure 4: Output of nslookup google.com showing resolved IPv4 and IPv6 addresses

Purpose:

Queries the DNS server to resolve a domain name to its corresponding IP address.

3.5 arp -a



```
C:\Users\Next>arp -a

Interface: 192.168.18.89 --- 0x5
    Internet Address      Physical Address      Type
    192.168.18.1          6c-14-6e-f4-5a-98    dynamic
    192.168.18.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.254.1 --- 0x9
    Internet Address      Physical Address      Type
    192.168.254.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static

Interface: 192.168.219.1 --- 0x14
    Internet Address      Physical Address      Type
    192.168.219.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
```

Figure 5: Output of arp -a showing the ARP table with IP to MAC address mappings

Purpose:

Displays the ARP table, showing the mapping between IP addresses and MAC addresses of devices recently communicated with on the local network.

4. Task 3: Router vs Switch

Comparison of a router and a switch across six key points.

Point	Router	Switch
Function	Connects different networks and routes data between them	Connects devices within the same local network
OSI Layer	Operates at Layer 3, Network Layer	Operates at Layer 2, Data Link Layer
Addressing Used	Uses IP addresses to forward data	Uses MAC addresses to forward data
Data Unit	Forwards packets	Forwards frames
Broadcast Domain	Separates broadcast domains	Single broadcast domain unless VLANs used
Common Use Case	Connecting a home or office network to the internet	Connecting multiple devices within one LAN, such as office PCs

5. Task 4: IPv4 vs IPv6

Comparison of IPv4 and IPv6 addressing across seven key points, including one example address for each.

Point	IPv4	IPv6
Address Length	32 bit	128 bit
Address Format	Decimal, separated by dots, e.g. 192.168.1.1	Hexadecimal, separated by colons, e.g. 2001:0db8:85a3:0000:0000:8a2e:0370:7334
Total Addresses	About 4.3 billion	Virtually unlimited, 340 undecillion
Address Configuration	Manual or DHCP	Supports auto-configuration (SLAAC) plus DHCPv6
Header Complexity	More complex, includes checksum	Simplified header, no checksum, faster processing
Security	Security not built in, relies on IPsec as add-on	IPsec support built into the protocol
NAT Requirement	Commonly requires NAT due to address shortage	Designed to remove the need for NAT

6. Task 5: OSI Model

All seven layers of the OSI Model, presented as a sequential stack from Layer 7 down to Layer 1, each with its primary function, a commonly used protocol, and an associated networking device.

LAYER 7	Application Layer Function: Provides network services directly to end user applications, such as browsers and email clients Protocol: HTTP Device: Firewall (application aware)
LAYER 6	Presentation Layer Function: Translates, encrypts, and compresses data so it is usable by the application layer Protocol: SSL/TLS Device: Gateway
LAYER 5	Session Layer Function: Establishes, manages, and terminates communication sessions between applications Protocol: NetBIOS Device: Gateway
LAYER 4	Transport Layer Function: Provides reliable end to end data delivery, error checking, and flow control Protocol: TCP Device: Firewall
LAYER 3	Network Layer Function: Handles logical addressing and routing of packets between different networks Protocol: IP Device: Router
LAYER 2	Data Link Layer Function: Handles physical addressing and reliable node to node data transfer within the same network Protocol: Ethernet Device: Switch
LAYER 1	Physical Layer Function: Transmits raw bit streams over the physical medium, such as cables or radio signals Protocol: Ethernet (cabling) Device: Hub / Network Cable

7. Challenges Faced

Initial confusion in identifying the correct active network adapter since multiple adapters (Ethernet, Wi-Fi, virtual) appeared in the ipconfig /all output. Resolved by cross-checking which adapter had an active IPv4 address and DHCP enabled. Tracert results also varied slightly in response time due to network congestion, resolved by re-running the command for consistent results.

8. Conclusion

This assignment strengthened foundational networking knowledge, including identifying network configuration details, using essential diagnostic commands, differentiating routers from switches, comparing IPv4 and IPv6 addressing, and understanding the structure and function of the OSI Model.

9. References

- <https://www.infosectrain.com/blog/top-20-networking-commands>
- <https://www.geeksforgeeks.org/computer-networks/differences-between-ipv4-and-ipv6/>